

Vertical distribution of foliar biomass in western larch (*Larix occidentalis*)

Geoffrey M. Williams, Andrew S. Nelson, and David L.R. Affleck

Abstract: Western larch (*Larix occidentalis* Nutt.) is an endemic pioneer species in northwestern North America and unique as a deciduous conifer and the most shade-intolerant, fastest growing, and most fire-resistant species in the northwestern United States. To better understand its production ecology, we used a multilevel modeling approach to analyze the intrinsic dynamics of western larch vertical foliage distribution and compared it with other species. We found that western larch allocates foliage into a more diffuse distribution as the crown lengthens, whereas shade-tolerant evergreens concentrate foliage into a more monolayered distribution higher within the crown as it lengthens. Crown foliar biomass scaled linearly with diameter at breast height, indicating that western larch does not fill volume in the crown with foliage at an increasing rate like other conifers. Our model supports the hypothesis that foliar shade intolerance and water stress jointly influence foliage allocation in this deciduous conifer. These results also highlight intrinsic foliage distribution as a factor potentially contributing to the inability of western larch to survive light-limiting conditions and its preference for mesic sites. The models developed here provide a basic framework that may be built upon to study the morphological response of western larch to modified stand conditions such as disturbance and silvicultural treatment.

Key words: biomass, foliage distribution, allometry, probability density function, western larch.

Résumé : Le mélèze occidental (*Larix occidentalis* Nutt.) est une espèce pionnière endémique du nord-ouest de l'Amérique du Nord et est unique au nord-ouest des États-Unis en étant à la fois un conifère décadu et l'espèce la plus intolérante à l'ombre, la plus résistante au feu et dont la croissance est la plus rapide. Afin de mieux comprendre l'écologie de sa production forestière, nous avons utilisé une approche de modélisation multiniveaux pour analyser la dynamique intrinsèque de la distribution verticale du feuillage du mélèze occidental et la comparer à celle d'autres espèces. Nous avons trouvé que la distribution du feuillage du mélèze occidental est de plus en plus diffuse à mesure que la cime s'allonge alors que les conifères tolérants à l'ombre concentrent leur feuillage selon une distribution plus monostratifiée plus haut dans la cime à mesure que celle-ci s'allonge. La biomasse foliaire de la cime était linéairement reliée au diamètre à hauteur de poitrine, ce qui indique que le mélèze occidental ne remplit pas le volume de la cime avec du feuillage à un taux croissant comme d'autres conifères. Notre modèle supporte l'hypothèse que l'intolérance à l'ombre du feuillage et le stress hydrique ont une influence conjointe sur l'allocation du feuillage de ce conifère décadu. Ces résultats soulignent aussi que la distribution intrinsèque du feuillage est un facteur qui contribue potentiellement à la préférence du mélèze pour les stations mésiques et à son incapacité à survivre dans des conditions de faible lumière. Les modèles mis au point dans la présente étude fournissent un cadre de base dont on peut tirer parti pour étudier la réaction morphologique du mélèze occidental en fonction de modifications de la condition du peuplement comme une perturbation ou un traitement sylvicole. [Traduit par la Rédaction]

Mots-clés : biomasse, distribution du feuillage, allométrie, fonction de densité de probabilité, mélèze occidental.

1. Introduction

Characterization of the vertical distribution of foliage in the crown is essential to understanding ontogenic changes and the underlying response of trees to modified stand conditions. The vertical distribution of foliage is the primary factor determining the distribution of photosynthetically active light intercepted by foliage (Horn 1971; Stenberg et al. 1994; Vose et al. 1995) and thus has a strong influence on growth. Therefore, understanding the differences in the response of vertical foliage distributions across species to changes in intrinsic (e.g., diameter, height, and age) and extrinsic (e.g., stand density, social position, and disturbance) variables will facilitate the optimization of management strategies. To enrich

general knowledge of forest ecology, a mechanistic framework to explain observations may then be proposed by relating contrasting foliage allocation strategies back to species traits such as relative shade or drought tolerance.

Vertical foliage distributions are also useful in applied forestry. Morphological changes provide a useful substitute for long-term growth responses to management (Buckley et al. 1997). Changes in crown morphology and leaf distribution can be used to predict growth response. For example, growth efficiency of lodgepole pine (*Pinus contorta* Dougl. ex Loud.) defined by stem increment per unit height per unit foliage was inversely proportional to crown length (Smith and Long 1989). However, crown length provides

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only a crude surrogate for foliar density and no specific information about foliage distribution.

The character and degree of plasticity of tree species' vertical foliage distributions to light and hydraulic conditions define their ability to respond to disturbance, management, and competition (Oliver and Larson 1990). Effects of treatments, environmental factors, intrinsic traits, and their interactions on vertical foliage distributions may be inferred through analysis of empirical models (Jerez et al. 2005; Garber and Maguire 2005). Probability density functions (PDFs) provide good approximations of vertical foliage distributions. PDFs may be fit to foliage measurements such as leaf area or biomass across absolute or relative dimensions such as distance along the main stem from the base of the crown toward the stem apex. Using a PDF framework, empirical vertical foliage distribution models may be further parameterized with intrinsic explanatory variables such as crown length, tree height, the ratio of crown length to height, and age (Maguire and Bennett 1996; Utsugi et al. 2006; Weiskittel et al. 2009).

Using the PDF framework, the form and characteristic dynamics of vertical foliage distributions have been found to vary across shade-tolerance levels, across hardwood species, between hardwoods and conifers, and within species by genotype (Nelson et al. 2014, 2015). Vertical foliage distributions peak at a lower relative stratum within the crown in shade-tolerant conifers relative to shade-intolerant species (Mäkelä and Vanninen 2001; Goudie et al. 2016). Vertical foliage distributions also change with respect to extrinsic factors such as social position (Xu and Harrington 1998; Kantola and Mäkelä 2004). The relative magnitude and direction of the shift in vertical foliage distribution in response to changing growing conditions may be influenced by leaf anatomy, capacity for drought tolerance, level of epinastic control, and other ecophysiological characters (Horn 1971; Oliver and Larson 1990).

North American larches (*Larix* spp.) are unique among sympatric conifers for their annually deciduous foliage, indeterminate annual growth, and high rate of epicormic branch replacement (Gower and Richards 1990; Lanner 1995), which together should allow for a high level of temporal plasticity in biomass allocation within the crown. Previous studies of vertical distribution of biomass in *Larix* spp. have been limited to the whole-canopy level or to wood biomass and did not measure crown foliage distribution (Bidlake and Black 1989; Li 2004). However, *Larix* spp. exhibit substantial spatial variation in foliage density, and previous studies suggest plasticity of biomass allocation to immediate growing conditions, especially light availability (Kurachi et al. 1986; Osawa 1990; Zhao and Wang 2004). In addition to contributing to general knowledge, the development of baseline models of vertical foliage distribution in the crown of *Larix* spp. are needed to provide a framework to study its response for the optimization of silvicultural treatments.

Western larch (*Larix occidentalis* Nutt.) has the highest wood density and is the fastest-growing and most fire-resistant conifer species in the northwestern United States (Schmidt and Shearer 1990; Kretschmann 2010). *Larix occidentalis* prefers mesic sites, especially drainages and northern aspects. Decline of *L. occidentalis* has been linked to fire suppression and climate change (Arno et al. 1997; Rehfeldt and Jaquish 2010). In addition to making it an ideal timber species, the high wood density and rapid growth of *L. occidentalis* also provide the ecological services of carbon storage and wildlife habitat. Dead *L. occidentalis* snags are rot-resistant and preferred over other species by cavity-nesting birds (McClelland et al. 1979). For these reasons, improved knowledge of the production ecology of *L. occidentalis* is tantamount to the sustainability of the cultural, economic, and ecological values of northwestern forests and particularly to the long-term goal of restoration of more natural fire regimes to the region.

Here we employ a multilevel modeling approach to characterize the behavior of vertical foliage distribution in *L. occidentalis* with respect to intrinsic factors. The specific objectives of the

Fig. 1. Native range (Little 1971) and sites where *Larix occidentalis* were harvested for the study.



study were to (i) develop allometric models for foliar mass at the branch and tree levels, (ii) evaluate the fit of PDFs to vertical distribution of foliage, and (iii) characterize how vertical foliage distributions change with respect to intrinsic variables. By developing a basic understanding of foliar biomass distribution within *L. occidentalis* crowns across a wide range of tree sizes and site conditions, the present study provides a strong modeling framework for optimizing *L. occidentalis* management strategies. We also explore how the foliage allocation morphospace of this unique species might be related to its autecology and physiology.

2. Materials and methods

2.1. Study site and data collection

A total of 64 *L. occidentalis* trees were destructively sampled from 20 locations across the northern Rocky Mountains in northern Idaho and western Montana between 2011 and 2016 (Fig. 1). Mean average temperature at the sites ranged from 3.8 °C to 9.3 °C, and mean annual precipitation ranged from 312 mm to 837 mm. Elevation ranged from 957 m to 2051 m, and latitude ranged from 44°45'13"N to 48°52'58"N.

Diameter at breast height (DBH), diameter at the base of the live crown (DBLC), total height (TH), and vertical distance along the stem from the base of the bole to the base of the live crown (HBLC) were measured. For every branch on every tree, the vertical distance along the stem from the base of the bole to the center of the branch at the junction with the stem (HIC) and the diameter of the branch within approximately 5 cm of the junction of the branch and the stem (BD) were measured. Additional tree- and branch-level variables were calculated, including branch cross-sectional area (BA) and relative positions of the branches within the crown (Table 1).

A subset of between one and eight branches per tree were randomly selected for measurement of total branch foliage biomass (Table 2). Branch segments were placed in an oven and dried with one of two regimens: segments from the Carscallen, TMU, and UI sites were dried for 72 h at 65 °C, and those from the rest of the sites were dried at 105 °C until constant mass was reached. After moisture mass was eliminated, foliage was removed from each segment and weighed separately on a balance.

2.2. Branch foliar mass models

Foliar mass measurements of 312 branches from 64 trees (Table 2) were used to fit models to estimate foliar mass as a function of branch- and tree-level variables. A range of equation forms and combinations of predictive variables have been employed for estimation of branch foliage mass (FM), branch leaf

Table 1. Variables tested to fit branch and tree level foliage models.

Variable	Definition	Formula
Branch		
ω_{ij} (tree)	Random effect of tree j within site i on branch	
BD (mm)	Branch diameter at stem junction	
BA (mm ²)	Branch cross sectional area at stem junction	$\pi BD^2/4$
HIC (cm)	Height into crown from crown base	
DIC (cm)	Depth into crown from stem apex	CL – HIC
RHIC	Relative height into crown from base	HIC/CL
RDIC	Relative depth into crown from apex	DIC/CL
FM (g)	Total foliar mass of branch	
Tree		
φ_i (site)	Random effect of site location i on tree	
TH (cm)	Tree height	
HBLC (cm)	Height from bole to base of crown	
CL (cm)	Length of the crown	TH – HBLC
DBLC (mm)	Diameter of stem at base of crown	
DBH (mm)	Diameter at breast height (1.3 m)	
CLR	Crown length ratio	CL/TH
MCLR	Modified crown ratio	CL/(TH – 1.3 m)
HMC (cm)	Height to the midpoint of the crown	HBLC + CL/2
CFM (g)	Total foliar mass in the tree crown	

area (LA), and branch woody biomass (BM) in past studies (Table 3). In addition to previous models used to predict FM, model forms previously used to estimate LA and BM were also adapted and tested for their ability to estimate FM in *L. occidentalis*. Variation due to unmeasured tree-level factors such as age and social position were accounted for by including random effects for site and tree within site (Garber and Maguire 2005; Nelson et al. 2014; Weiskittel et al. 2009).

In addition to previously published models, modified and original models were also fit to the branch FM data. The modified and original models were obtained by substituting equivalent variables (e.g., RDIC for RHIC) or by sequentially adding variables to base models. Variables that significantly improved model fit were retained. Models were fit and parameters were recovered using maximum likelihood estimation with the nlme function in the nlme package of R (R Core Team 2017; Pinheiro et al. 2017), which allows incorporation of hierarchical random effects and the ability to account for autocorrelation and unequal variance. Starting parameter values specified in the nlme function were first obtained by least squares estimation from the PROC MODEL function in SAS (2016; SAS Institute Inc., Cary, North Carolina).

Parameter estimates were screened for significance using a threshold p value of 0.05, and nonsignificant parameters were considered to be zero and removed from the model. Once a set of models with all significant parameters was obtained, the relative explanatory power, or goodness of fit, for each model was assessed using multiple selection criteria: likelihood ratio tests, adjusted and generalized R^2 , the Akaike information criterion (AIC), and root mean squared error (RMSE). Among the best-fitting models with statistically equivalent performance across fit statistics, the simplest model that showed a significant improvement in goodness of fit over a base model was retained and selected for estimation of foliage mass for the rest of the branches.

2.3. Crown foliar mass models

The branch summation method (Monserud and Marshall 1999) was employed to obtain an estimate of total crown foliar mass. Of the original 64 trees, 14 were not included because they did not have height measurements for all of the branches. Using the branch model with fixed effects only, foliar mass was estimated for 7408 individual branches and summed across each of the remaining 50 tree crowns (Appendix Table A1). When model fitting

and selection were repeated on branch foliage estimates that incorporated random effects, the same model was selected with minimal differences in parameter estimates compared with estimates obtained using only fixed effects.

A variety of previously published (Table 4), modified, and original linear and nonlinear mixed effects models were fit to the crown foliar mass estimates to test the significance of intrinsic tree variables (Table 1) by examining the p values of their parameters. The same model selection criteria employed to select the best branch foliage mass models were used to select the best crown foliar mass model. Variation due to unmeasured site-level environmental variables and stand conditions were accounted for by including a random effect for site. Models were fit using maximum likelihood estimation with the nlme function in R (R Core Team 2017; Pinheiro et al. 2017) from starting parameter values obtained from PROC MODEL in SAS (2016; SAS Institute Inc.).

2.4. Vertical distribution of foliage in the crown

Of the 50 trees for which foliage was estimated across all of the branches, one was excluded as an outlier from the data used in vertical foliage distribution model testing because it only had 10 branches in the crown. Vertical distribution of FM estimated with the branch foliage mass model using only fixed effects for all of the branches in the crowns of the remaining 49 *L. occidentalis* trees (Table 5) were used to fit the left- and right-truncated (eq. 1) and untruncated two-parameter Weibull, the β distribution (eq. 2), and the Johnson's S_B (eq. 3) functions. The same vertical distribution PDF and parameter models were selected when random effects were used to estimate branch foliage. The general forms of the PDFs are expressed as

$$(1) \quad p(X) = \frac{\beta X^{\beta-1} e^{-[(X/\eta)^\beta - (\gamma/\eta)^\beta]}}{\eta^\beta}$$

$$(2) \quad p(X) = \frac{X^c(1-X)^{d-1}}{\Gamma(c)\Gamma(d)/\Gamma(c+d)}$$

$$(3) \quad p(X) = \frac{\tau e^{-(1/2)(\psi + \tau \ln[X/(1-X)])^2}}{\sqrt{2\pi X(1-X)}}$$

where X represents the position in the crown, and $p(X)$ is the relative density of foliar mass at that position such that $p(X)dX$ is the proportion of total crown FM over an interval. Relative depth in the crown (RDIC; Table 1) was used for X to facilitate ease of comparison across crowns of different lengths (Weiskittel et al. 2009; Schneider et al. 2011; Nelson et al. 2014). Absolute values may be obtained by rescaling the interval of the fitted PDF from $[0, 1]$ to the measured crown length of each tree and (or) rescaling $p(X)$ by total crown FM (Maguire and Bennett 1996; Garber and Maguire 2005). In the Weibull function (eq. 1), β is the skew (shape) parameter, η is the kurtosis (scale) parameter, and γ is the truncation point. In the β distribution (eq. 2), c and d are shape parameters, and $\Gamma(x)$ is the gamma function. In Johnson's S_B (eq. 3), ψ and τ are shape parameters.

Relative to using a continuous value of RDIC for X , the “binning” approach (e.g., Stenberg et al. 1993; Garber and Maguire 2005; Weiskittel et al. 2009) smooths data and improves the fit of PDFs during model fitting. Aggregating foliage into vertical strata of equal length reduces noise associated with variation in the size of branches. To improve fitting of PDFs, foliage was partitioned into 20 equal segments along the length of the stem in each crown.

Root mean square error (RMSE) and mean absolute bias (MAB) were calculated from observed data and predicted values summed across the bins for each tree. The distribution of error and bias across tree size was examined by plotting error and absolute bias against DBH. RMSE and overall MAB were also calculated across

Table 2. Branches sampled for branch foliar mass model.

Site	No. of trees	Branches		BD (mm)			RHIC			FM (g)		
		Total no.	No. per tree	Min	Mean	Max	Min	Mean	Max	Min	Mean	Max
Bandy	5	19	3.8	8.00	35.11	69.00	0.07	0.34	0.60	4.00	91.48	246.04
Carscallen	6	36	6.0	3.82	10.78	24.38	0.06	0.48	0.87	0.57	38.02	148.47
Coram 2	1	8	8.0	7.00	13.25	17.00	0.17	0.25	0.41	4.50	19.91	37.50
Deer Creek	1	2	2.0	13.00	17.00	21.00	0.15	0.21	0.26	25.00	26.50	28.00
Fortine Sinclair	1	6	6.0	15.00	26.67	50.00	0.39	0.58	0.77	26.00	61.67	164.00
G-Man	2	10	5.0	7.00	22.00	31.00	0.15	0.40	0.62	3.00	73.80	132.00
Lubrecht Old Coloma	1	4	4.0	36.00	45.25	58.00	0.12	0.18	0.29	40.00	134.80	232.95
Lubrecht River Flats	11	47	4.3	5.00	29.49	63.00	0.01	0.40	0.76	2.10	104.38	486.00
McCall 1	2	13	6.5	20.32	41.42	58.42	0.15	0.42	0.86	30.50	201.23	435.00
McCall 3	3	11	3.7	20.32	36.02	50.80	0.29	0.56	0.82	51.50	186.55	317.00
McCall 4	5	25	5.0	7.62	33.22	63.50	0.16	0.48	0.84	6.00	185.88	833.50
McCall 5	1	4	4.0	15.24	19.68	22.86	0.42	0.59	0.77	55.00	67.88	78.50
Ninemile Edith Peak	2	9	4.5	10.00	17.89	25.00	0.39	0.54	0.67	11.00	43.56	88.00
Ninemile Petty Creek	3	19	6.3	7.00	33.63	45.00	0.20	0.43	0.71	10.00	146.05	310.00
Rexford Pinkham	7	25	3.6	6.00	13.92	50.00	0.05	0.34	0.62	2.00	23.01	265.19
Seeley Morrell Creek	2	13	6.5	14.00	34.77	51.00	0.28	0.52	0.84	17.00	94.12	194.45
St Maries	2	7	3.5	15.00	29.29	35.00	0.09	0.33	0.56	6.00	70.83	249.23
TMU	3	18	6.0	3.82	11.06	17.96	0.11	0.50	0.85	2.53	30.89	116.64
UI217	4	24	6.0	5.09	11.87	20.56	0.05	0.50	0.84	5.70	34.40	78.50
UI486	2	12	6.0	4.78	9.19	17.08	0.14	0.50	0.81	4.19	17.42	45.52
All	64	312	4.9	3.82	23.67	69.00	0.01	0.44	0.87	0.57	84.97	833.50

Note: Min, minimum; Max, maximum.

Table 3. Previously published models used to estimate total branch foliage.

Species ^a	Variable ^b	Class ^c	Branch model formula	Citation
PISY	FM	Quad	$BA(a_0RDIC + a_1RDIC^2 + a_2) + a_3$	Mäkelä and Vanninen 2001
TSCA	LA	Other	$(a_0 + a_1\sqrt[3]{BA} + a_2\sqrt[3]{RDIC})^3$	Kenefic and Seymour 1999
PITA	FM	Power	$BD^{a_1}DIC^{a_2}$	Gillespie et al. 1994
Multiple	LA/BM/FM	Power	$a_0DIC^{a_1}BA^{a_3}$	Monserud and Marshall 1999
POTR	FM/BM	Power	$a_0BD^{a_1}DIC^{a_2}\left(\frac{TH}{BD}\right)^{a_3}$	Ek 1979
PISY	FM	Power	$a_0RDIC^{a_1}RHIC^{a_2}BA^{a_3}$	Mäkelä and Vanninen 2001
PITA	FM	Mixed	$a_0BD^{a_1}e^{a_2RHIC^{a_3}}$	Xu and Harrington 1998
PIRU	LA	Mixed	$e^{a_0BD^{a_1}RDIC^{a_2}}e^{a_3RDIC}$	Maguire et al. 1998
PSME	LA/FM	Mixed	$a_0BD^{a_1}RDIC^{a_2-1}e^{a_3RDIC^{a_4}}$	Maguire and Bennett 1996
ABGR PICO	LA	Mixed	$a_0BD^{a_1}RDIC^{a_2-1}e^{-(a_3RDIC)^{a_4}}$	Garber and Maguire 2005
PIAB	FM/BM	Mixed	$RDIC^{a_1}BA^{a_2}e^{a_3RDIC^{a_4}}e^{a_5}$	Kantola and Mäkelä 2004
PIPO	LA	Mixed	$(a_0 - \varphi_1)BD^{a_1}RDIC^{a_2-1}e^{-(a_3RDIC)^{a_4}}$	Garber and Maguire 2005
Multiple	LA	Mixed	$BD^{a_1}RDIC^{a_2-1}e^{-(a_3+\varphi_1)RDIC^{a_4}}$	Nelson et al. 2014 ^d
Multiple	LA	Mixed	$a_0BD^{a_1}RDIC^{a_2-1}e^{[(a_3+\varphi_1+\omega_1)RDIC^{a_4}]^2}$	Weiskittel et al. 2009

^aPSME, *Pseudotsuga menzeisii* var. *menzeisii*; PITA, *Pinus taeda*; ABGR, *Abies grandis*; PICO, *Pinus contorta*; PIPO, *Pinus ponderosa*; POTR, *Populusx Tristis*; PIAB, *Picea abies*; PISY, *Pinus sylvestris*; TSCA, *Tsuga canadensis*.

^bLA, leaf area; FM, foliar biomass; BM, branch biomass.

^cPower, power series; Quad, quadratic; Mixed, mixed power series and exponential.

^dNelson et al. 2014 accounted for branch angle in the original model, but *L. occidentalis* has near 90° branch angles, so we present a model where RDIC replaces distance to branch tip and to start of foliage on the branch in the power and exponential portions of the original model, respectively.

trees and used as model selection criteria to select the best-fitting PDF. Shape parameters were recovered for each tree from each PDF by maximum likelihood estimation (Weiskittel et al. 2009; Nelson et al. 2014). Tree vertical foliage distribution shape parameter estimates for the best-fitting PDF were then modeled as functions of tree variables to discern how the distribution changed with respect to intrinsic factors (Table 1).

Several model forms have been used to estimate parameters for vertical distribution of foliage in previous studies (Table 6). During model selection, published, modified, and original equation forms were tested, and the simplest model (fewest parameters) among models that showed equivalent performance across a range of fit statistics was selected. Variation due to unmeasured site-level variables was accounted for by including random effects for site. Models were fit and their parameters were estimated by

weighted least squares with fixed effects using the nls function and by maximum likelihood with mixed effects using the nlme function in R (R Core Team 2017; Pinheiro et al. 2017). Starting parameter values were obtained from the PROC MODEL function in SAS (2016; SAS Institute Inc.).

The vertical foliage distribution model was simulated on representative trees from the dataset. To simulate the model by height class, representative trees were obtained by partitioning the dataset into three quartiles by TH centered around the 25th percentile, the median, and the 75th percentile and taking median values from within each partition for the other variables. To simulate the model by crown length, the dataset was broken into the top, middle, and bottom deciles by tree height. Representative trees within each height class were then obtained by calculating median values of TH, DBLC, and minimum, median, and maximum

Table 4. Previously published models used to estimate crown foliage.

Species ^a	Variable ^b	Class ^c	Crown model formula	Citation
TSCA	LA	Mixed	$b_0\text{DBH}^{b_1+b_2\text{CL}+b_3(\text{TH}/\text{DBH})}$	Weiskittel et al. 2009
PSME	FM	Mixed	$b_0\text{CL}^{b_1}e^{b_2\text{DBH}/\text{TH}}$	Maguire and Bennett 1996
Multiple	LA	Mixed	$b_0\text{DBH}^{b_1}e^{(b_2+\varphi_1)(\text{DBH}/\text{CL})}$	Nelson et al. 2014
PSME	FM/LA/BM	Power	$b_0\text{DBH}^{b_1}\text{CLR}^{b_2}\text{CL}^{b_3}$	Monserud and Marshall 1999 ^d
PIPO				
PIMO				
TSCA	LA	Linear	$b_0 + b_1(\text{DBH} + \text{MCLR})$	Kenefic and Seymour 1999 ^e
CHOB	LA	Power	$b_0\text{DBH}^{b_1}$	Utsugi et al. 2006
PTTA	LA	Power	$\frac{1}{\text{HMC}}\left(\frac{\text{DBH}}{b_1}\right)^{b_2}$	Jerez et al. 2005

^aPSME, *Pseudotsuga menzeisii*; PTTA, *Pinus taeda*; PIPO, *Pinus ponderosa*; PIMO, *Pinus monticola*; TSCA, *Tsuga canadensis*; CHOB, *Chamaecyparis obtusa*.
^bLA, leaf area; FM, foliar biomass; BM, branch biomass.
^cFunction classes: Power, power series; Mixed, mixed power series and exponential.
^dModified from the original published version that also included sapwood area and crown competition factor.
^eOriginal regression was weighted by DBH.

Table 5. Trees used to fit vertical foliar distribution models.

Site	No. of trees	DBH (mm)			TH (cm)			DBLC (mm)			CL (cm)		
		Min	Mean	Max	Min	Mean	Max	Min	Mean	Max	Min	Mean	Max
Bandy	2	214.0	251.5	289.0	1849	1953	2056	170.0	196.5	223.0	1250	1257	1264
Carscallen	6	61.0	115.6	162.6	652	944	1137	71.1	149.9	246.4	585	879	1094
Coram 2	1	159.0	159.0	159.0	1894	1894	1894	94.0	94.0	94.0	827	827	827
Fortine Sinclair	1	317.0	317.0	317.0	2564	2564	2564	200.0	200.0	200.0	1307	1307	1307
Lubrecht River Flats	3	305.0	390.0	466.0	2887	2913	2954	220.0	255.0	309.0	2000	2089	2161
McCall 1	2	360.7	422.9	485.1	2993	3094	3194	233.7	265.4	297.2	1430	1599	1768
McCall 3	3	269.2	400.5	495.3	2478	2858	3072	208.3	312.4	388.6	1503	1681	1905
McCall 4	5	185.4	351.5	607.1	1948	2590	3423	81.3	268.2	533.4	951	1620	2359
McCall 5	1	264.2	264.2	264.2	2185	2185	2185	218.4	218.4	218.4	1722	1722	1722
Ninemile Edith Peak	2	164.0	174.5	185.0	1781	1997	2212	128.0	140.0	152.0	1397	1473	1549
Ninemile Petty Creek	3	307.0	388.0	443.0	2266	2481	2761	163.0	262.3	318.0	1354	1594	1762
Rexford Pinkham	7	118.0	170.6	206.0	1158	1573	1777	82.0	111.0	136.0	644	920	1201
Seeley Morrell Creek	2	380.0	394.0	408.0	3008	3011	3013	286.0	294.5	303.0	2263	2505	2746
St Maries	2	255.0	285.0	315.0	2490	2492	2493	134.0	144.5	155.0	742	924	1106
Tower Mixup	3	76.2	125.3	188.0	811	1079	1277	76.2	125.3	188.0	649	930	1140
UI217	4	96.5	137.2	177.8	817	1058	1295	99.1	141.0	167.6	683	908	1055
UI486	2	73.7	101.6	129.5	762	1012	1262	71.1	100.3	129.5	634	843	1052
All	49	61.0	247.0	607.1	652	1951	3423	71.1	188.6	533.4	585	1294	2746

Note: Min, minimum; Max, maximum.

Table 6. Previously published models used to estimate foliage distribution parameters.

Species ^a	Variable ^b	Function	Parameter model formulas ^c	Citation
TSCA	FM/LA	Beta	$c = g_0 + g_1\text{CFM}$ $d = g_2\text{TH}^{g_3}e^{g_3\text{CR}+g_4\text{TH}}$	Maguire and Bennett 1996
Multiple	LA	Weibull	$\eta = g_0\text{CL}^{g_1+g_2\text{CLA}}$ $\beta = g_3\text{CL}^{g_4+g_5\text{CLA}+g_6\text{TH}}$	Weiskittel et al. 2009
CHOB	LA	Weibull	$\eta = g_0 + g_1\text{CLA} + g_2\text{TH}$ $\beta = g_3 + g_4\text{CLA} + g_5\text{TH}$	Utsugi et al. 2006

^aTSCA, *Tsuga canadensis*; CHOB, *Chamaecyparis obtusa*.
^bLA, leaf area; FM, foliar biomass.
^cCLA, total crown leaf area; CFM, total crown foliar biomass.

CL for each decile. Within the middle tree height decile, mean CFM estimated by branch summation from eq. 4 was used, and within the top and bottom deciles, median estimated CFM was used.

3. Results

3.1. Branch foliar mass model

A total of 91 branch FM models of a variety of forms borrowed and adapted from the literature were tested and compared (Appendix Table A2). A modified form of the Maguire and Bennett (1996) model (eq. 4) had the highest log likelihood score, the lowest AIC and RMSE, and the highest generalized and adjusted R² for

both fixed and combined fixed and random effects (Table 7). It was selected as the best model. Equation 4 includes random effects for site and tree within site in both the BD and RHIC terms and a constant in the RHIC term (Fig. 2). The model form was

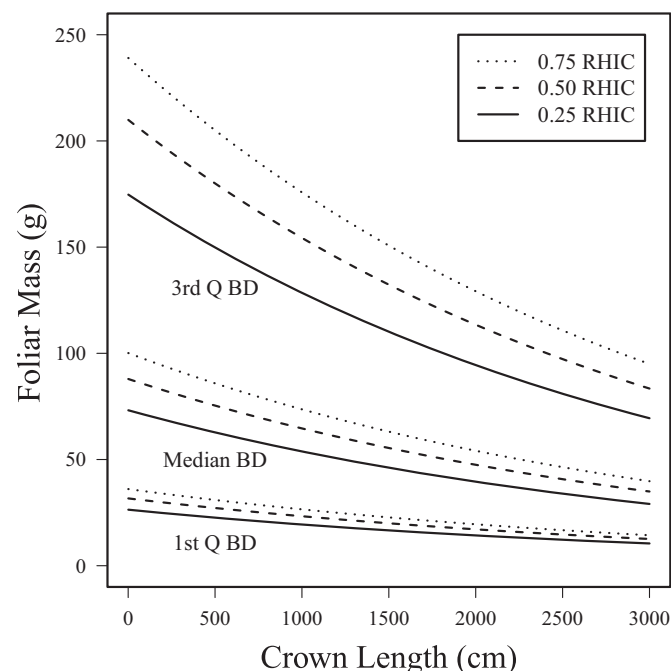
(4)
$$\text{FM} = a_0\text{BD}^{a_1+\varphi_i+\omega_{ij}}(0.25 + \text{RHIC})^{a_2+\varphi'_i+\omega'_{ij}-1}e^{a_3\text{CL}}$$

where a_0 , a_1 , a_2 , and a_3 are parameters and φ_i , φ'_i , ω_{ij} , and ω'_{ij} are random effects from site i on BD, from site i on RHIC, from tree j within site i on BD, and from tree j within site i on RHIC, respec-

Table 7. Fit statistics for models selected to estimate FM and its distribution.

Eq.	Model	LL	AIC	RMSE	Adjusted R ²		Generalized R ²	
					Fixed	Fixed + random	Fixed	Fixed + random
4	Branch FM	-1653.52	3329.03	42.830	0.634	0.847	0.638	0.849
5	Crown FM	-419.26	850.52	1032.050	0.832	0.934	0.835	0.935
6	Kurtosis (η)	70.52	-131.04	0.057	0.516	0.516	0.536	0.536
7	Skew (β)	-41.18	92.35	0.497	0.522	0.731	0.542	0.742

Note: LL, log likelihood; AIC, Akaike information criterion; RMSE, root mean square error.

Fig. 2. Branch foliar mass predicted by eq. 4 by crown length (CL), relative height in the crown (RHIC), and branch diameter quartile (BD). Q, quartile.

tively. When applied to estimate foliar mass, the original form of the selected model (eq. 8 in Appendix Table A2) predicted zero foliar mass for the lowermost branches in the crowns. Because crown length was defined by live branches, an additional parameter was added to the relative height value before it was raised to its exponent to correct for the zero FM estimate obtained for the lowest branches. The additional parameter was estimated to be 0.25 but was not significant ($p > 0.05$). To keep all parameters significant, the actual value of 0.25 was substituted in eq. 4 as a constant. Substitution of a constant did not change the number of parameters and therefore could not be tested in a log likelihood ratio test for a significant improvement in model fit; however, the log likelihood improved ($\delta = 0.96$). The exponential term for crown length significantly improved model fit over a base model that included only BD and RHIC (eq. 8 over eq. 15, Appendix Table A2) in a log likelihood ratio test ($\chi^2 = 4.76$, $df = 1$, $0.025 < p < 0.05$). RHIC had more explanatory power than RDIC. Substituting RHIC for RDIC in the base model conferred a marginal reduction in error and increase in R^2 and a substantial improvement in log likelihood ($\delta = 3.56$; eq. 18 over eq. 15, Appendix Table A2). Random effects contributed to the R^2 of eq. 4 (Table 7).

3.2. Crown foliar mass model

In total, 26 different models for total crown foliage of a variety of forms original or borrowed and adapted from the literature were tested and compared (Appendix Table A3). Of the models tested with all significant parameters, the highest log likelihood, lowest AIC, and highest generalized and adjusted R^2 for fixed

effects (Table 7) were obtained from a modified form of the Kenefic and Seymour (1999) model for total crown foliage, with random effects for site incorporated into coefficients for both DBH and HMC. The model form was

$$(5) \quad CFM = (b_1 + \varphi_i'')DBH + (b_2 + \varphi_i''')HMC$$

where b_1 and b_2 are parameters and φ_i'' and φ_i''' are random effects from site i on DBH and HMC, respectively. In a simple linear model, absolute residuals from eq. 5 were positively correlated with DBH partitioned into 5-inch bins ($F = 9.499$, $df_1 = 1$, $df_2 = 48$, $p = 0.0034$, $R^2 = 0.1652$). Like the original Kenefic and Seymour (1999) model (eq. 115 in Appendix Table A3), eq. 5 expresses total crown foliage as a linear function of DBH (Fig. 3; Fig. 4), but unlike the original model, it includes HMC instead of MCLR. There was a large improvement in the likelihood between the original and modified forms ($\delta = 7.77$). Overall, the lowest RMSE and highest generalized and adjusted R^2 for combined fixed and random effects were obtained from an original form that was a mixed function with a power term for DBH and an exponential CL term (eq. 114 in Appendix Table A3), and inclusion of CL improved model fit over a base model (eq. 121 in Appendix Table A3) in a likelihood ratio test ($\chi^2 = 5.98$, $df = 1$, $p < 0.015$). However, eq. 5 was retained for superior R^2 for fixed effects. Inclusion of random effects improved the R^2 (Table 7).

3.3. Vertical distribution of foliage in the crown

When fitting distribution models to the estimates of branch foliage for the branches of 49 trees (Appendix Tables A1 and A2), the right-truncated two-parameter Weibull gave the lowest RMSE and MAB of the five PDFs tested (Table 8). Both whole-tree RMSE and MAB of the Weibull estimates increased steadily with DBH (Fig. 5). Across all trees, the maximum likelihood parameter estimates for the right-truncated two-parameter Weibull had a mean of 0.684 ± 0.085 and range of 0.448 to 0.858 for kurtosis (η), and a mean of 3.475 ± 0.917 and range of 2.064 to 6.362 for skew (β). Overall, the majority of foliage was located in the lower crown. Peak foliage for the average values of the kurtosis and skew parameters occurred at RDIC = 0.62.

A total of 27 different models of a variety of forms were fit to the kurtosis (η) parameter estimates and compared (Appendix Table A4). The best-fitting model of the kurtosis parameter was a linear function of CL and DBLC and had the form

$$(6) \quad \eta = g_0 + g_1CL + g_2DBLC$$

where g_0 , g_1 , and g_2 are parameters. In a likelihood ratio test, eq. 6 showed significant improvement over a model expressing kurtosis as linear function of CL alone (eq. 139 in Appendix Table A4; $\chi^2 = 6.19$, $df = 1$, $0.01 < p < 0.025$). Random effects for site did not improve fit and were not used (eq. 124 in Appendix Table A4). Equation 6 had the highest log likelihood, lowest AIC and RMSE, and highest generalized and adjusted fixed R^2 among models tested, with the exception of a marginally better fitting exponentially transformed linear model that also included CFM (eq. 123 in Appendix Table A4). Because inclusion of CFM reduced fit in the

Fig. 3. Crown foliar mass predicted by eq. 5 by diameter at breast height (DBH) for quartiles of height to middle of the crown (HMC). Q, quartile; Med, median.

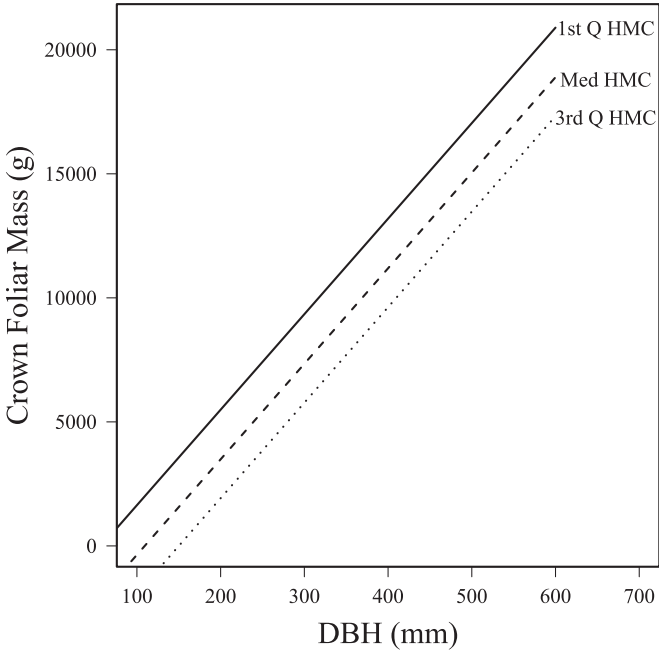


Fig. 4. Predicted and actual* crown foliar mass by diameter at breast height (DBH)[†] for *Larix occidentalis* compared with predictions of models for other species (western white pine, Monserud and Marshall 1999; balsam fir and red spruce, Marchand 1984; subalpine fir, Smith and Long 1989; and lodgepole pine, Long and Smith 1989). *Actual values depicted as circles were estimated by branch summation from eq. 4. [†]Height to the middle of the crown for eq. 5 was estimated using the linear model HMC = 316.8 + 3.9976DBH ($F = 157.6$, $df_1 = 1$, $df_2 = 47$, $p < 2.2 \times 10^{-16}$, $R^2 = 0.7702$).

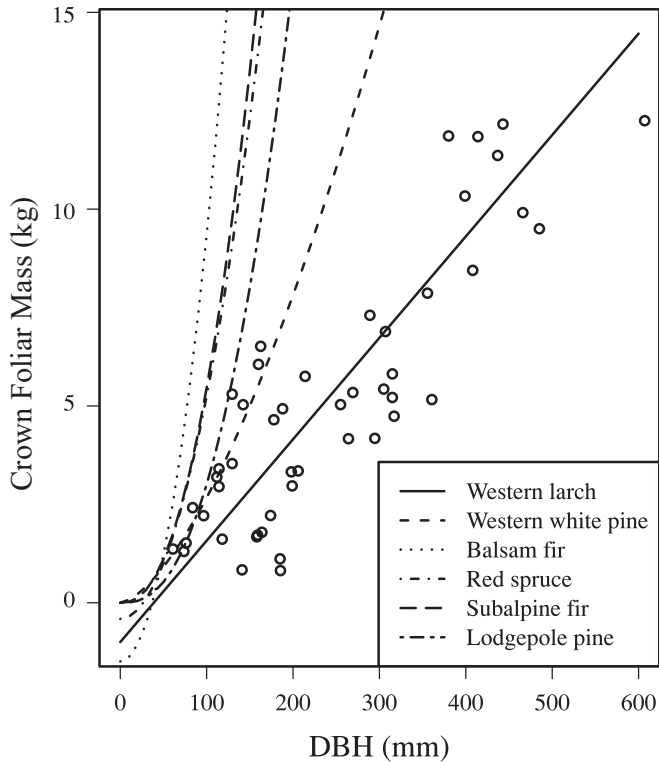
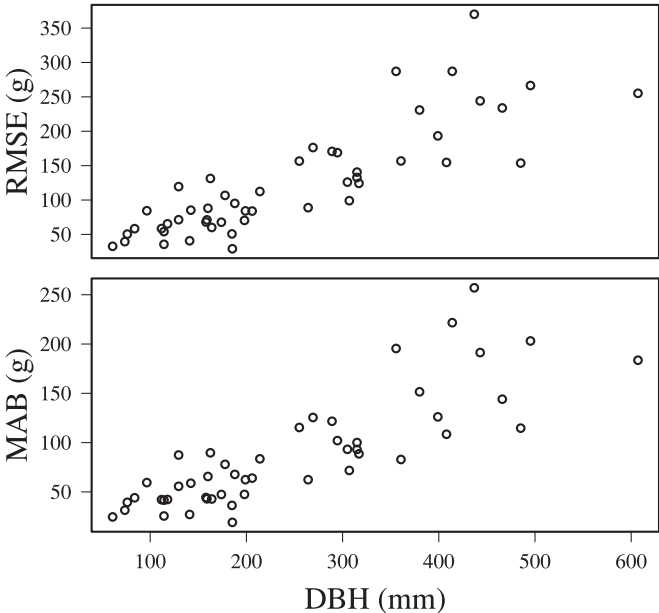


Table 8. Fit statistics for PDFs.

Model	RMSE (g)	MAB (g)
Weibull (right-truncated)	125.16	88.26
Weibull (no truncation)	126.17	88.84
Beta	134.53	95.68
Weibull (left-truncated)	135.86	95.21
Johnson's S_B	157.15	112.86

Note: RMSE, root mean square error; MAB, mean absolute bias.

Fig. 5. Root mean square error (RMSE) and mean absolute bias (MAB) by tree size (diameter at breast height, DBH) of Weibull predictions of relative foliar density in each bin.



linear version of eq. 123 (eq. 125 in Appendix Table A4), eq. 6 was retained for the sake of simplicity.

Eighteen models were fit to the skew (β) parameter estimates (Appendix Table A5). A power function of CLR with CFM in the exponent (estimated from eq. 4 without random effects) was selected:

(7)
$$\beta = g_3 \text{CLR}^{g_4 + \varphi_i''''} + g_5 \text{CFM}$$

where g_3 , g_4 , and g_5 are parameters and φ_i'''' is the random effect from site i on CLR. A higher log likelihood, lower AIC and RMSE, and higher generalized and adjusted R^2 for combined fixed and random effects were obtained by including a random effect with CFM (eq. 149 in Appendix Table A5); however, eq. 149 was rejected for its inferior R^2 for fixed effects. Equation 7 was a better fit than a base model univariate power function of CLR (eq. 156 in Appendix Table A5) in a likelihood ratio test ($\chi^2 = 8.09$, $df = 1$, $p < 0.005$). Inclusion of random effects improved the R^2 of eq. 7 (Table 7). When simulating the model on representative trees, foliage became most concentrated in the median height class and shifted upward with increasing crown length (Figs. 6–7). Based on parameter estimates for eqs. 6–7 (Table 9), kurtosis (η) increases as DBLC increases and as CL decreases. Skew (β) increases as CFM increases and as CLR decreases. Substituting $\text{CLR} = \text{CL}/\text{TH}$, skew decreases as TH increases when CL is constant and decreases as CL increases when TH is constant (Fig. 8).

Table 9. Parameter estimates for models of vertical foliage distribution kurtosis and skew.

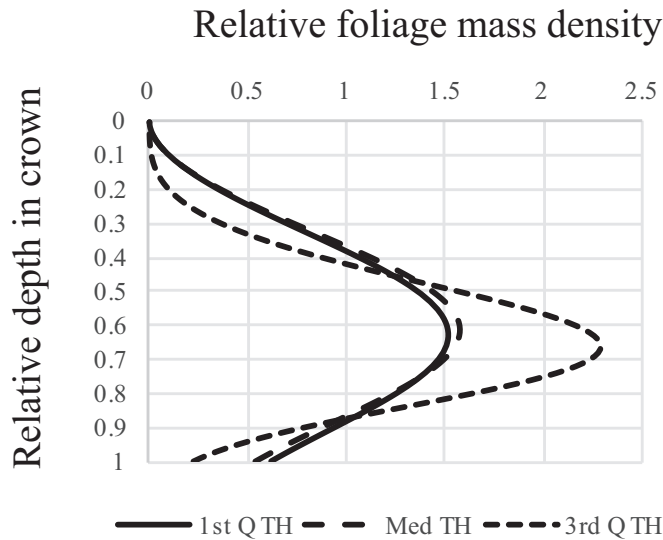
Model	Parameter	Estimate	SE fixed ^a	SD random ^b (site)	p
Weibull kurtosis (η), eq. 6	g_0	0.818712	0.023273	NA ^c	0.000000
	g_1	-0.000204	0.000030	NA	0.000000
	g_2	0.000686	0.000162	NA	0.000195
Weibull skew (β), eq. 7	g_3	2.632306	0.127060	NA	0.000000
	g_4	-0.883330	0.130914	0.181416	0.000000
	g_5	0.000043	0.000015	NA	0.008123

^aStandard errors for fixed effects.

^bStandard deviations for random effects.

^cNA, model did not include random effects.

Fig. 6. Predicted vertical foliar mass distribution for representative *Larix occidentalis* trees by height class under the model: 1st quartile (Q), TH = 893.1 cm, CL = 819.9 cm, CFM = 2684 g, DBLC = 118.1 mm; Median (Med), TH = 1190 cm, CL = 1053 cm, CFM = 7989 g, DBLC = 156.2 mm; 3rd Q, TH = 1758 cm, CL = 955 cm, CFM = 2216.9 g, DBLC = 115.2 mm.



4. Discussion

4.1. Branch foliar mass

A combination of both branch- and tree-level variables provided the best predictive power to estimate branch foliar mass. Equation 4 resembles several branch foliage models selected for other species, but with notable exceptions. In previous models for the shade-tolerant *Picea rubens* Sarg. and the shade-intolerant *Pinus contorta* and *Pinus ponderosa* Dougl. ex Laws., branch leaf area increased with increasing depth into the crown (Maguire et al. 1998; Garber and Maguire 2005). In the shade-intolerant *Pinus sylvestris* L., whether branch foliar mass increased or decreased with branch depth was site-specific (Mäkelä and Vanninen 2001), suggesting the ability to adapt to light conditions. In the shade-intolerant *Pinus taeda* L., branch foliar mass increased with increasing relative height in the crown (Gillespie et al. 1994; Xu and Harrington 1998), as it does in *Larix occidentalis* according to eq. 4. The light compensation point for photosynthesis is higher in shade-intolerant species such as *P. taeda* and *L. occidentalis*. Thus, lower branches support less foliage because they are shaded by upper branches and unable to fix enough carbon to meet local respiratory demand.

Due to the exponential term in the model, branch foliar mass also depended on crown length. Whereas branch foliar mass in-

Fig. 7. Predicted vertical foliar mass distribution for representative *Larix occidentalis* trees of minimum, median and maximum crown length (CL) within (a) top, (b) middle, and (c) bottom deciles by height (TH). (a) TH = 3072 cm, DBLC = 340.4 mm, CFM = 11 860 g, minimum (Min) CL = 1634 cm, median (Med) CL = 1905 cm, and maximum (Max) CL = 2359 cm. (b) TH = 1894 cm, DBLC = 128 mm, CFM = 3475.9 g, Min CL = 827 cm, Med CL = 1250 cm, and Max CL = 1549 cm. (c) TH = 810.8 cm, DBLC = 76.2 mm, CFM = 1518 g, Min CL = 585.2 cm, Med CL = 649.2 cm, and Max CL = 798.6 cm.

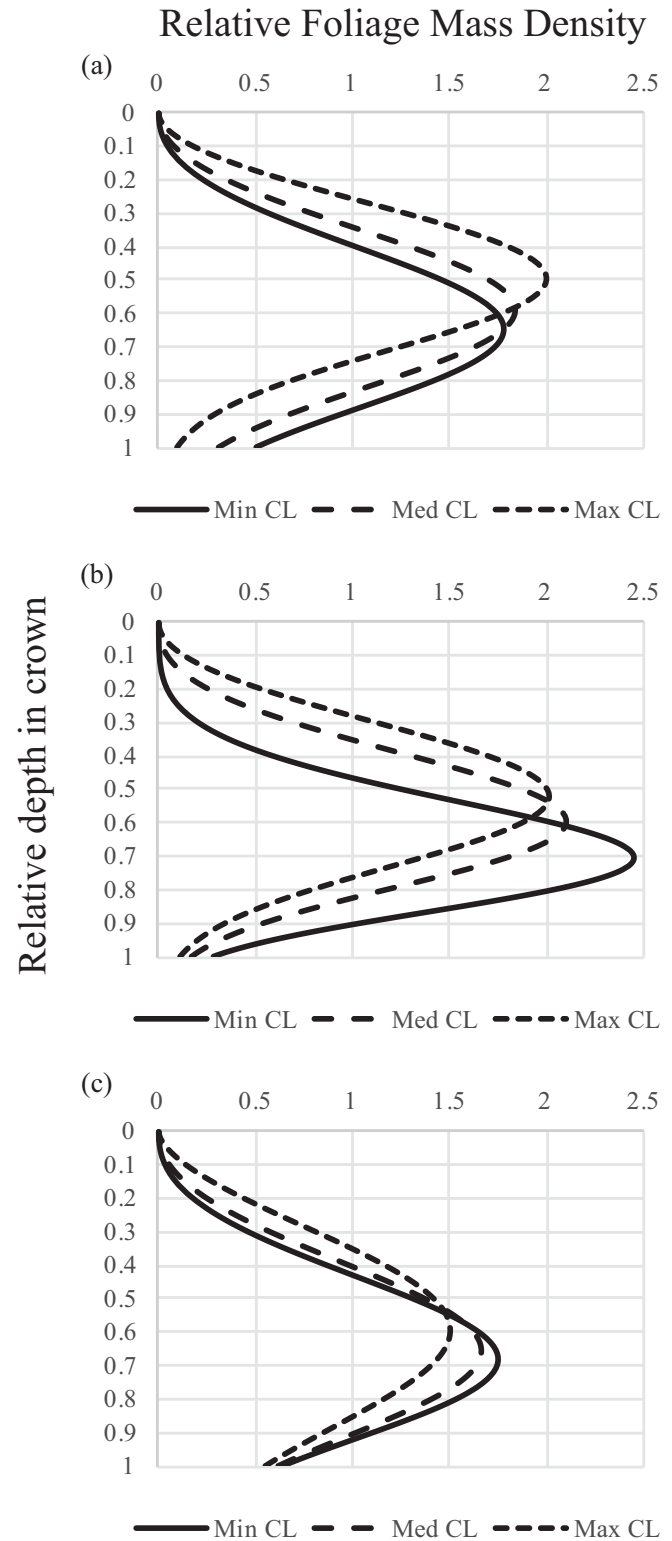


Table 10. Parameter estimates for branch and crown foliar mass models.

Model	Parameter	Estimate	SE fixed ^a	SD random ^b		
				Site	Tree in site	<i>p</i>
Branch	<i>a</i> ₀	0.602347	0.194738	NA ^c	NA	0.002210
	FM, <i>a</i> ₁	1.706984	0.101006	0.356257	0.102046	0.000000
	eq. 4 <i>a</i> ₂	1.452303	0.144354	0.217986	0.451876	0.000000
	<i>a</i> ₃	−0.000308	0.000126	NA	NA	0.015335
Crown	<i>b</i> ₁	38.380153	3.522708	5.924948	NA	0.000000
	FM, <i>b</i> ₂	−3.155260	0.588623	0.449660	NA	0.000008
	eq. 5					

^aStandard errors for fixed effects.
^bStandard deviations for random effects.
^cNA, model did not include random effects.

creased positively with branch diameter and relative height, branches of similar diameter and relative height had less foliage in longer crowned trees than shorter crowned trees (Fig. 2), most likely due to self-shading. Because *L. occidentalis* replaces all of its foliage with a fresh cohort every year, where foliage is produced and how much can be supported by a branch may both be plastic to immediate light conditions. Equation 4 is simpler than models for other species, which often include a second exponential term for relative branch position (Table 3) (Maguire et al. 1998; Xu and Harrington 1998; Kantola and Mäkelä 2004; Garber and Maguire 2005; Weiskittel et al. 2009; Nelson et al. 2014). Less plasticity in branch foliage production to absolute branch position and greater plasticity to overall crown size further indicate greater dependence of branch-level foliage biomass production on light conditions relative to hydraulic or structural constraints.

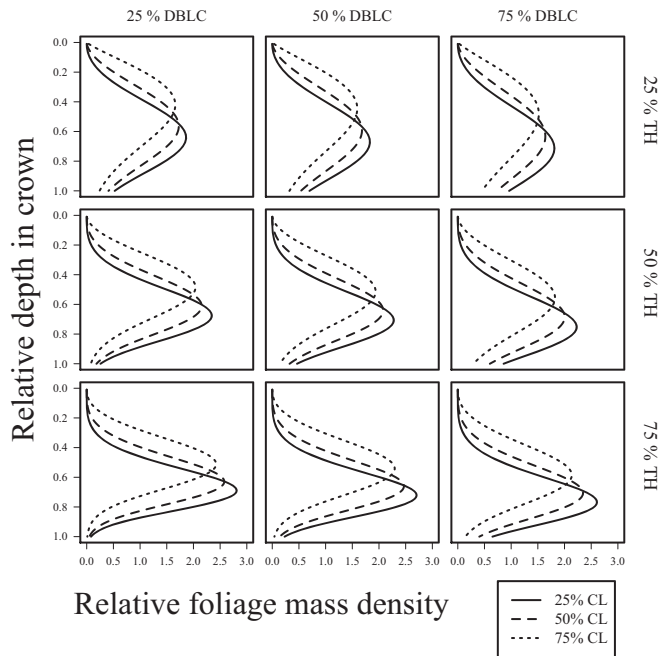
4.2. Crown foliar mass

A linear model fit crown foliar mass for *L. occidentalis* as well as or better than nonlinear models (Appendix Table A3). Because the range of tree sizes sampled for this study was within the range of previous studies, sampling scheme can be ruled out as a factor contributing to the selection of a linear over a nonlinear form. Kenefic and Seymour (1999) published a linear model to predict crown foliar mass from DBH and modified crown length ratio in *Tsuga canadensis* (L.) Carrière. Ultimately, height to the middle of the crown was chosen as a superior substitute for crown length ratio (for relationships, see Table 1). Cross-sectional area and height to the middle of the crown were also used to model total crown leaf area for *P. contorta* and *Abies lasiocarpa* (Hook.) Nutt. but fit a power series rather than a linear model (Smith and Long 1989; Long and Smith 1989).

Crown foliar mass was negatively correlated with height to the middle of the crown in *L. occidentalis* (Table 10; Fig. 3). Using the relationship $HMC = TH - 0.5CL$, this finding agrees with consistent observations across species that crown mass increases with crown length but decreases with increasing height (Dean and Long 1986; Maguire and Bennett 1996; Weiskittel et al. 2009; Monserud and Marshall 1999), although most previous models were nonlinear. Crown foliar mass was also positively correlated with DBH but inversely related to height to middle of the crown in *P. taeda* (Jerez et al. 2005). Unlike deciduous conifers, evergreens accumulate more foliage by retaining cohorts from previous years, which may lead to higher observed rates of foliage biomass scaling (Fig. 4) (Marchand 1984; Long and Smith 1989; Smith and Long 1989; Monserud and Marshall 1999). The greater variance in crown foliar mass among older trees could be due to accumulated deterioration from disease and severe weather events.

In contrast, in seedlings with small DBH and height to the middle of the crown, crown foliar mass might increase as fast or faster in *L. occidentalis* than in other species. Evergreens with crown foliage mass that scales as a power or exponential function of DBH

Fig. 8. Predicted vertical distribution of foliar mass in *Larix occidentalis* at median CFM and quartiles of measured values of diameter at crown base (DBLC), tree height (TH), and crown length (CL), using the relationship $CLR = CL/TH$ (Table 1).



should accumulate foliage at a slower rate than *L. occidentalis* when young if the initial slope of the exponential crown foliage model is less than the slope of the linear crown foliage model (Fig. 4). The estimated linear coefficient for DBH in the linear crown foliar mass model for *L. occidentalis* (Table 7) is almost twice as high as the initial slope parameter estimate for allometric power series models of crown foliar mass for *P. ponderosa* and nearly eight times higher than the initial slope for *Pinus monticola* Dougl. ex D. Don in Idaho (Monserud and Marshall 1999).

Besides being inconsistent with models for other species, the linear relationships between crown foliar mass, DBH, and height to the middle of the crown in *L. occidentalis* do not support the theory of allometric scaling of foliage with a power function of diameter across tree species life histories and niche space (Osawa et al. 1991; Enquist et al. 1999). However, linear scaling of crown foliar mass with DBH and height to the middle of the crown does support the hypothesis that foliar crown volume filling is limited by self-shading. *Larix occidentalis* is among the most shade-intolerant species of conifers (Gower and Richards 1990) and its crown foliage scales with body mass with a fractal dimension outside the distribution observed across conifer species (Duursma et al. 2010).

4.3. Vertical distribution of foliage in the crown

As expected based on previous findings (Maguire and Bennett 1996; Weiskittel et al. 2009), the vertical distribution of foliage in the crown depended on crown length, crown length ratio, and crown foliar mass. The positive correlation of DBH with RMSE and MAB (Fig. 5) is likely due to larger absolute predicted and actual values of foliar mass per vertical bin in larger trees, because total crown foliar mass was positively correlated with DBH. Older trees also had a longer period of potential exposure to extrinsic factors such as shade and belowground competition during stand development, as well as a higher cumulative probability of experiencing severe weather, pests, disease, etc. Extrinsic factors could have contributed to greater variation in the distribution that could not be accounted for with our model.

Consistent with the exponential term for crown length in the branch foliage models used to generate foliage estimates for individual branches, crown length and crown length ratio had significant effects on the vertical distribution of foliage in *L. occidentalis*. The downward shift and much more monolayered foliage distribution predicted by our model for taller *L. occidentalis* (Table 9; Figs. 6–8) are similar to those of other shade-intolerant species (Van Hees and Bartelink 1993; Maguire and Bennett 1996). With respect to crown length, *L. occidentalis* has a more diffuse distribution of foliage centered at a higher position in longer crowned trees relative to shorter crowned trees (Fig. 8), which is consistent with the hypothesis that self-shading acts as an important constraint on foliage allocation in this species. However, the tallest trees showed the opposite trend — a more concentrated distribution of foliage as the crown lengthened — when the model was simulated on representative trees generated from summary statistics (Fig. 7). Similar to these findings, *Pseudotsuga menziesii* var. *menziesii* (Mirb.) Franco vertical foliage distribution shows increasing kurtosis and an upward shift under conditions of increased shade competition (Maguire and Bennett 1996). In contrast to *L. occidentalis* and *P. menziesii*, shade-tolerant evergreens shift foliage upward into a more monolayered distribution with increasing crown length even at young ages (Weiskittel et al. 2009).

The preferential allocation of foliage within a more condensed stratum lower in the crowns of taller shade-intolerant trees may reflect a balancing act between minimizing water loss while maximizing light capture. Matrix and gravitational water potential are more negative higher in the canopy, (Pallardy and Kozlowski 2010), potentially limiting foliage production. Vapor deficit is likely to be higher in the upper canopy of tall trees and may exert more influence than stem water deficit on stomatal behavior in *Larix* spp. (Higgins et al. 1987; Anfodillo et al. 1998). Soil water content may not be an important factor for *L. occidentalis* rooted deeply on moist sites, but low humidity occurs frequently in the west. By allocating foliage in a monolayer deep in the crowns of dominant and codominant trees, the stomata can remain open, and high levels of photosynthesis may be maintained as long as there is an ample supply of water from the stem. It is also possible that the tallest trees in our dataset were of dominant social position such that the lower portion of the crown received ample light, allowing more foliage to be supported on the lower branches.

Most other studies found that some measure of total crown foliage significantly accounted for variance in its vertical distribution (Maguire and Bennett 1996; Utsugi et al. 2006; Weiskittel et al. 2009). This widespread observation makes sense because greater amounts of total foliage are likely to lead to greater self-shading and hydraulic constraints on the vertical foliage distribution. However, unlike previous studies, our model of vertical foliage distribution also included diameter at the base of the crown (DBLC) as a significant component of the variance in the vertical foliage distribution. The slightly more upward-shifted and more diffuse distribution of foliage associated with smaller values of DBLC under our model for *L. occidentalis* (Table 9; Fig. 8) could be the consequence of reduced diameter growth in dense stands due to resource competition and increased amounts of side shade.

5. Conclusions

Larix occidentalis appears to exhibit dynamic patterns of biomass allocation and retention driven by immediate growing conditions. Preferential allocation of biomass to better-lit portions of the canopy drives the observed variation in foliar mass and its distribution. Shorter crowns support more foliage per branch and higher densities of foliage in better-lit strata. Notably, the representative trees from the 1st and 3rd quartiles by height both had longer crowns than the representative tree of median height.

These observations may be due to branch shedding in trees of the median height class in response to competition with neighboring trees, in contrast to crown lengthening under dominant social position for tall trees and open site conditions for the youngest trees. As a result, trees of intermediate height exhibited a more leptokurtic foliage distribution than the smallest or largest trees (Figs. 6 and 7). The younger trees in the dataset had many more branches than the older trees, indicating self-pruning as the stand develops and side shade increases.

The plasticity of the vertical foliage distribution of *L. occidentalis* to light availability is likely facilitated by higher turnover of biomass and indeterminate development, in line with previously observed spatial correlation between biomass production and light intensity within the crowns of other *Larix* spp. (Kurachi et al. 1986; Osawa 1990). In another shade-intolerant conifer (*Pinus banksiana* Lamb.), observed variation in vertical distribution of foliage, though markedly opposite *L. occidentalis* with respect to crown length and height, could be accounted for primarily by variation in allocation of internodal foliage (Schneider et al. 2011). Shade intolerance has a physiological basis at the leaf level such that new foliage produced in the spring is passively allocated to well-lit and more productive portions of the crown.

The intrinsic dynamics of vertical foliage distribution in *L. occidentalis* may arise from the interaction between light and hydraulic factors. Unlike broadleaf deciduous trees, especially shade-tolerant hardwoods, which concentrate foliage in a monolayer at the top of the crown, *L. occidentalis* foliage has a relatively diffuse distribution more similar to its coniferous relatives, with peak density closer to the crown base (Maguire and Bennett 1996; Weiskittel et al. 2009). A multilayered crown form is inferior in the understory but hydraulically advantageous on the exposed, disturbed sites where *L. occidentalis* prefers to regenerate (Horn 1971). Because *L. occidentalis* foliage is relatively unprotected, constraints from more negative water potentials and (or) higher vapor deficits in the upper crown may impose some limit on the magnitude of the upward shift in the foliage distribution in response to shade. However, this study did not explicitly investigate the effect of light levels, water potential, social position, or stand density on vertical foliage allocation. Also, due to the unbalanced study design, various extrinsic factors influencing foliage allocation may have been confounded with study plot differences in stand condition and (or) age.

Larix occidentalis often produces narrow, highly receded crowns in codominant situations. Our model and basic knowledge of larch autecology suggest that these typical morphological characteristics are symptomatic of suboptimal growth and resource limitation. We suggest that managers optimize stand density to facilitate healthy crown development and carefully monitor crown form to maximize growth.

Acknowledgements

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Appendix A

Appendix Tables appear on following pages.

Table A1. Summary statistics for branches used to predict foliar mass using eq. 4 to estimate crown foliar mass and to fit vertical foliage distribution models.

Site	No. of branches	BD				RHIC		
		Min	Mean	SD	Max	Mean	SD	Max
Bandy	172	2.00	13.83	11.09	47.00	0.54	0.21	0.84
Bandy	171	2.00	13.68	7.00	29.00	0.50	0.20	0.80
Carscallen	343	1.52	5.71	2.70	19.92	0.52	0.24	0.94
Carscallen	343	2.92	8.22	3.84	23.72	0.50	0.26	0.95
Carscallen	384	2.21	7.66	3.26	18.91	0.53	0.26	0.96
Carscallen	159	1.80	5.67	2.28	11.98	0.48	0.27	0.92
Carscallen	297	2.94	10.30	6.06	60.80	0.51	0.26	0.97
Carscallen	217	1.49	6.62	3.03	15.08	0.50	0.25	0.91
Coram 2	56	6.00	12.64	3.76	22.00	0.41	0.12	0.57
Fortine Sinclair	91	6.00	17.88	7.99	36.00	0.51	0.22	0.87
Lubrecht River Flats	132	4.00	24.64	15.64	70.00	0.55	0.21	0.86
Lubrecht River Flats	232	2.00	11.06	7.86	45.00	0.60	0.22	0.87
Lubrecht River Flats	133	4.00	23.67	10.44	45.00	0.70	0.19	0.93
McCall 1	59	7.62	25.31	15.23	55.88	0.28	0.15	0.51
McCall 1	121	7.62	22.31	14.91	63.50	0.61	0.23	0.92
McCall 3	102	7.62	17.88	8.26	38.10	0.51	0.19	0.75
McCall 3	93	2.54	31.65	18.62	88.90	0.50	0.18	0.73
McCall 3	121	7.62	32.77	12.02	58.42	0.52	0.21	0.86
McCall 4	75	7.62	19.14	9.61	43.18	0.44	0.20	0.70
McCall 4	100	7.62	17.63	9.21	43.18	0.51	0.23	0.81
McCall 4	22	7.62	13.97	4.35	22.86	0.48	0.13	0.63
McCall 4	82	7.62	28.22	11.72	76.20	0.43	0.18	0.65
McCall 4	118	7.62	30.39	21.05	86.36	0.53	0.25	0.91
McCall 5	102	7.62	15.91	7.62	60.96	0.58	0.21	0.84
Ninemile Edith Peak	86	2.00	10.24	5.78	25.00	0.47	0.22	0.77
Ninemile Edith Peak	39	5.00	11.90	6.51	25.00	0.46	0.24	0.78
Ninemile Petty Creek	136	6.00	25.89	11.20	49.00	0.48	0.24	0.88
Ninemile Petty Creek	133	6.00	25.36	13.01	52.00	0.56	0.24	0.92
Ninemile Petty Creek	99	7.00	21.14	8.38	43.00	0.54	0.20	0.84
Rexford Pinkham	84	6.00	10.19	1.94	14.00	0.36	0.16	0.61
Rexford Pinkham	10	6.00	8.60	1.35	10.00	0.13	0.08	0.22
Rexford Pinkham	93	5.00	13.23	4.73	24.00	0.55	0.17	0.78
Rexford Pinkham	91	5.00	14.28	5.18	25.00	0.49	0.18	0.73
Rexford Pinkham	85	7.00	11.80	2.44	18.00	0.42	0.17	0.65
Rexford Pinkham	32	5.00	11.78	3.08	20.00	0.32	0.14	0.51
Rexford Pinkham	61	6.00	12.05	3.31	20.00	0.40	0.16	0.60
Rexford Pinkham	105	4.00	12.90	4.60	25.00	0.51	0.17	0.72
Seeley Morrell Creek	181	4.00	19.81	11.27	48.00	0.68	0.23	0.93
Seeley Morrell Creek	123	6.00	28.62	13.40	56.00	0.65	0.22	0.90
St Maries	61	5.00	21.48	9.58	38.00	0.59	0.20	0.81
St Maries	77	6.00	20.64	9.15	48.00	0.45	0.18	0.72
TMU	351	1.84	6.09	2.89	17.81	0.52	0.25	0.95
TMU	251	1.91	9.56	5.55	24.72	0.57	0.25	0.96
TMU	121	1.63	6.87	3.10	16.17	0.58	0.27	0.96
UI217	394	1.76	8.17	4.34	23.68	0.56	0.25	0.98
UI217	176	2.22	6.87	2.86	15.53	0.53	0.25	0.95
UI217	225	1.99	7.93	4.01	22.64	0.53	0.25	0.93
UI217	253	1.61	8.87	5.72	23.26	0.56	0.22	0.91
UI486	128	2.24	6.31	2.55	12.84	0.49	0.27	0.90
UI486	288	2.20	6.94	4.31	29.25	0.58	0.22	0.91
All	7408	1.49	13.09	10.88	88.90	0.53	0.24	0.98

Note: Min, minimum; Max, maximum; SD, standard deviation.

Table A2. Models evaluated for goodness of fit to branch foliar mass.

Branch foliar mass model equation		Rand	LL	AIC	RMSE	Adjusted R ²		Generalized R ²	
						Fixed	Fixed + random	Fixed	Fixed + random
(4)	$a0 * (BD^{a1}) * ((0.25 + RHIC)^{(a2 - 1)} * \exp(a3 * CL))$	a1 + a2	-1653.5	3329.0	42.83	0.634	0.847	0.638	0.849
(8)	$a0 * (BD^{a1}) * (RHIC^{(a2 - 1)} * \exp(a3 * CL))$	a1 + a2	-1654.5	3330.9	43.04	0.631	0.845	0.634	0.847
(9)	$(BD^{a1}) * (RHIC^{(a2 - 1)} * (CL^{a3}) * (HBLC^{a4}))$	a1 + a2	-1655.3	3332.6	43.31	0.612	0.841	0.615	0.843
(10)	$(BD^{a1}) * (RHIC^{(a2 - 1)} * (CL^{a3}))$	a1 + a2	-1655.5	3331.1	43.33	0.625	0.842	0.628	0.843
(11)	$(BD^{a1}) * (RHIC^{(a2 - 1)} * (TH^{a3}))$	a1 + a2	-1656.1	3332.1	43.30	0.617	0.842	0.619	0.843
(12)	$(a0 * BD^{a1}) * (RHIC^{(a2 - 1)} * (HBLC^{a3}))$	a1 + a2	-1656.2	3334.3	42.92	0.589	0.847	0.593	0.848
(13)	$(a0 * BD^{a1}) * (RHIC^{(a2 - 1)} * \exp(a3 * HBLC))$	a1 + a2	-1656.7	3335.5	43.35	0.622	0.841	0.626	0.843
(14)	$(a0 * BD^{a1}) * (RHIC^{(a2 - 1)} * (DBLC^{a3}))$	a1 + a2	-1656.8	3335.7	43.41	0.619	0.841	0.623	0.843
(15)	$(a0 * BD^{a1}) * (RHIC^{(a2 - 1)})$	a1 + a2	-1656.9	3333.7	43.41	0.617	0.841	0.619	0.842
(16)	$(a0 * BA^{a1}) * (RHIC^{(a2 - 1)})$	a1 + a2	-1657.2	3334.4	43.43	0.616	0.841	0.619	0.842
(17)	$(a0 * BD^{a1}) * (RDIC^{(a2 - 1)}) * (TH^{a3})$	a1 + a2	-1657.8	3337.6	43.77	0.600	0.839	0.604	0.840
(18)	$(a0 * BD^{a1}) * (RDIC^{(a2 - 1)})$	a1 + a2	-1660.4	3340.8	43.77	0.608	0.840	0.611	0.841
(19)	$a0 * BA^{a1} * RHIC^{a2} * TH^{a3} * CL^{a4} * DBLC^{a5}$	a1	-1661.4	3338.8	44.59	0.602	0.830	0.608	0.832
(20)	$a0 * BA^{a1} * RHIC^{a2} * TH^{a3} * CL^{a4} * DBLC^{a5}$	a1	-1661.4	3338.8	44.59	0.602	0.830	0.608	0.832
(21)	$a0 * BD^{a1} * RHIC^{a2} * TH^{a3} * CL^{a4} * DBLC^{a5}$	a1	-1661.5	3339.1	44.61	0.602	0.830	0.609	0.832
(22)	$a0 * BA^{a1} * DIC^{a2} * (HBLC/BA)^{a3}$	a1	-1662.3	3336.6	44.63	0.599	0.831	0.603	0.832
(23)	$a0 * BA^{a1} * DIC^{a2} * (HBLC/BD)^{a3}$	a1	-1662.3	3336.6	44.63	0.599	0.831	0.603	0.832
(24)	$a0 * BD^{a1} * DIC^{a2} * (HBLC/BA)^{a3}$	a1	-1662.4	3336.8	44.65	0.599	0.831	0.603	0.832
(25)	$a0 * BA^{a1} * DIC^{a2} * (TH/BA)^{a3}$	a1	-1663.2	3338.4	44.91	0.618	0.828	0.621	0.830
(26)	$a0 * BA^{a1} * DIC^{a2} * (TH/BD)^{a3}$	a1	-1663.2	3338.4	44.91	0.618	0.828	0.621	0.830
(27)	$a0 * BD^{a1} * DIC^{a2} * (TH/BA)^{a3}$	a1	-1663.3	3338.6	44.92	0.618	0.828	0.621	0.830
(28) ^a	$a0 * BD^{a1} * DIC^{a2} * (TH/BD)^{a3}$	a1	-1663.3	3338.6	44.92	0.618	0.828	0.621	0.830
(29)	$a0 * BA^{a1} * RDIC^{a2} * (TH/BA)^{a3}$	a1	-1663.5	3339.1	44.76	0.600	0.830	0.604	0.832
(30)	$a0 * BA^{a1} * RDIC^{a2} * (TH/BD)^{a3}$	a1	-1663.5	3339.1	44.76	0.600	0.830	0.604	0.831
(31)	$a0 * BD^{a1} * RDIC^{a2} * (TH/BA)^{a3}$	a1	-1663.7	3339.3	44.77	0.600	0.830	0.604	0.831
(32)	$a0 * BA^{a1} * RDIC^{a2} * (HBLC/BA)^{a3}$	a1	-1663.8	3341.7	44.54	0.567	0.832	0.571	0.833
(33)	$a0 * BA^{a1} * RHIC^{a2} * (TH/BA)^{a3}$	a1	-1663.9	3339.8	44.84	0.602	0.829	0.606	0.831
(34)	$a0 * BA^{a1} * RHIC^{a2} * (TH/BA)^{a3}$	a1	-1663.9	3339.8	44.84	0.602	0.829	0.606	0.831
(35)	$a0 * BD^{a1} * RHIC^{a2} * (TH/BA)^{a3}$	a1	-1664.0	3340.0	44.85	0.602	0.829	0.606	0.831
(36)	$a0 * BA^{a1} * RDIC^{a2} * (HBLC/BA)^{a3}$	a1	-1664.5	3341.1	44.56	0.565	0.832	0.569	0.834
(37)	$a0 * BA^{a1} * RDIC^{a2} * (HBLC/BD)^{a3}$	a1	-1664.5	3341.1	44.56	0.565	0.832	0.569	0.834
(38)	$a0 * BD^{a1} * RDIC^{a2} * (HBLC/BA)^{a3}$	a1	-1664.7	3341.3	44.57	0.565	0.832	0.569	0.834
(39)	$a0 * BA^{a1} * RHIC^{a2} * (HBLC/BA)^{a3}$	a1	-1664.9	3341.8	44.61	0.564	0.832	0.569	0.833
(40)	$a0 * BA^{a1} * RHIC^{a2} * (HBLC/BA)^{a3}$	a1	-1664.9	3341.8	44.61	0.565	0.832	0.569	0.833
(41)	$a0 * BA^{a1} * HIC^{a2} * (TH/BA)^{a3}$	a1	-1664.9	3341.9	44.79	0.584	0.830	0.588	0.832
(42)	$a0 * BA^{a1} * HIC^{a2} * (TH/BA)^{a3}$	a1	-1664.9	3341.9	44.79	0.584	0.830	0.588	0.832
(43)	$a0 * BD^{a1} * RHIC^{a2} * (HBLC/BA)^{a3}$	a1	-1665.1	3342.1	44.63	0.565	0.832	0.569	0.833
(44)	$a0 * BD^{a1} * HIC^{a2} * (TH/BA)^{a3}$	a1	-1665.1	3342.1	44.80	0.584	0.830	0.588	0.832
(45)	$a0 * (HIC^{a1}) * (BA^{a2})$	a1 + a2	-1666.3	3352.7	44.08	0.596	0.839	0.599	0.840
(46)	$a0 * (HIC^{a1}) * (BD^{a2})$	a1 + a2	-1666.6	3353.2	44.09	0.596	0.839	0.599	0.840
(47)	$a0 * BA^{a1} * HIC^{a2} * (HBLC/BA)^{a3}$	a1	-1667.1	3346.2	44.67	0.535	0.832	0.540	0.833
(48)	$a0 * BA^{a1} * HIC^{a2} * (HBLC/BA)^{a3}$	a1	-1667.1	3346.2	44.67	0.535	0.832	0.540	0.833
(49)	$a0 * BD^{a1} * HIC^{a2} * (HBLC/BA)^{a3}$	a1	-1667.3	3346.5	44.69	0.536	0.832	0.540	0.833
(50)	$(BA^{a1}) * (RHIC^{(a2 - 1)})$	a1	-1668.7	3347.5	45.46	0.614	0.825	0.615	0.826
(51)	$a0 * (DIC^{a1}) * (BD^{a2})$	a1	-1668.9	3349.7	45.38	0.628	0.826	0.630	0.827
(52)	$a0 * (DIC^{a1}) * (BA^{a2})$	a1	-1668.9	3349.7	45.38	0.628	0.826	0.630	0.827
(53)	$BA^{a1} * RDIC^{a2} * TH^{a3} * DBLC^{a4}$	a4	-1668.9	3351.8	45.27	0.629	0.827	0.632	0.828
(54)	$(BD^{a1}) * (DIC^{a2})$	a2	-1669.7	3349.5	45.40	0.625	0.827	0.626	0.827
(55)	$a0 * (DIC^{a1}) * (BD^{a2})$	a1	-1669.9	3349.8	45.63	0.628	0.825	0.630	0.826
(56) ^b	$a0 * (DIC^{a1}) * (BA^{a2})$	a1	-1669.9	3349.8	45.63	0.628	0.825	0.630	0.826
(57)	$(a0 * BD^{a1}) * (RDIC^{a2}) * \exp(a3 * RHIC)$	a3	-1663.8	3341.7	46.19	0.622	0.817	0.626	0.819
(58)	$(a0 * BD^{a1}) * (RHIC^{a2}) * \exp(a3 * RHIC)$	a3	-1663.9	3339.8	46.21	0.619	0.817	0.623	0.819
(59)	$(a0 * BD^{a1}) * \exp(a2 * RHIC)$	a2	-1663.9	3339.8	46.23	0.620	0.818	0.623	0.819
(60)	$\exp(a0) * BD^{a1} * RHIC^{a2} * \exp(a3 * RHIC)$	a3	-1664.0	3340.0	46.36	0.618	0.816	0.622	0.818
(61)	$\exp(a0) * BA^{a1} * RHIC^{a2} * \exp(a3 * RHIC)$	a3	-1664.5	3341.1	46.36	0.618	0.816	0.622	0.818
(62)	$\exp(a0) * BA^{a1} * RDIC^{a2} * \exp(a3 * RHIC)$	a3	-1664.5	3341.1	46.37	0.621	0.816	0.625	0.818
(63)	$(a0 * BD^{a1}) * (\exp(a2 * (RHIC)))$	a2	-1664.7	3341.3	46.39	0.620	0.816	0.623	0.817
(64)	$(a0 * BA^{a1}) * (\exp(a2 * (RHIC)))$	a2	-1664.9	3341.8	46.39	0.620	0.816	0.623	0.817
(65) ^c	$(BD^{a1}) * (DIC^{a2})$	a2	-1664.9	3341.8	45.71	0.625	0.825	0.626	0.825
(66)	$(BA^{a1}) * (DIC^{a2})$	a2	-1664.9	3341.9	45.78	0.622	0.824	0.623	0.824
(67)	$BA^{a1} * RDIC^{a2}$	a1	-1664.9	3341.9	46.14	0.603	0.820	0.604	0.821
(68)	$a0 * (HIC^{a1}) * (BA^{a2})$	a1	-1665.1	3342.1	45.71	0.598	0.824	0.601	0.825
(69)	$a0 * (HIC^{a1}) * (BD^{a2})$	a1	-1665.1	3342.1	45.71	0.598	0.824	0.601	0.825
(70)	$(a0 * BD^{a1}) * (RDIC^{a2}) * (DBLC^{a3})$	a2	-1666.3	3352.7	47.31	0.607	0.807	0.611	0.809
(71)	$(a0 * BA^{a1}) * (RDIC^{a2 - 1})$	a2	-1666.6	3353.2	47.39	0.613	0.807	0.615	0.809

Table A2 (concluded).

Branch foliar mass model equation	Rand	LL	AIC	RMSE	Adjusted R ²		Generalized R ²	
					Fixed	Fixed + random	Fixed	Fixed + random
(72) $\exp(a_0) * BA^{a_1} * RDIC^{a_2} * \exp(a_3 * RDIC)$	a3	-1667.1	3346.2	47.24	0.617	0.812	0.621	0.814
(73) ^d $\exp(a_0) * BD^{a_1} * RDIC^{a_2} * \exp(a_3 * RDIC)$	a3	-1667.1	3346.2	47.24	0.617	0.812	0.621	0.814
(74) $(BA^{a_1}) * (HIC^{a_2})$	a2	-1667.3	3346.5	48.16	0.596	0.805	0.597	0.805
(75) $(BD^{a_1}) * (HIC^{a_2})$	a2	-1668.7	3347.5	48.55	0.592	0.802	0.593	0.802
(76) $\exp(a_0) * BA^{a_1} * RHIC^{a_2} * \exp(a_3 * RDIC)$	a3	-1668.9	3349.7	47.95	0.609	0.806	0.612	0.808
(77) $\exp(a_0) * BD^{a_1} * RHIC^{a_2} * \exp(a_3 * RDIC)$	a3	-1668.9	3349.7	47.95	0.609	0.806	0.612	0.808
(78) $(a_0 * BD^{a_1}) * (\exp(a_2 * (RDIC)))$	a2	-1668.9	3351.8	47.95	0.610	0.807	0.612	0.808
(79) $(a_0 * BA^{a_1}) * (\exp(a_2 * (RDIC)))$	a2	-1669.7	3349.5	47.95	0.610	0.807	0.612	0.808
(80) $(a_0 + a_1 * (BA^{(1/3)})) + a_2 * (RDIC^{(1/3)})^3$	a0	-1669.9	3349.8	48.58	0.603	0.804	0.605	0.805
(81) $(a_0 + a_1 * (BA^{(1/3)})) + a_2 * (RHIC^{(1/3)})^3$	a0	-1669.9	3349.8	48.66	0.608	0.803	0.610	0.805
(82) $(a_0 + a_1 * (BA^{(1/3)})) + a_2 * (RDIC^{(1/3)})^3$	a0	-1691.1	3392.3	48.69	0.604	0.803	0.606	0.804
(83) $(a_0 + a_1 * (BD^{(1/3)})) + a_2 * (RDIC^{(1/3)})^3$	a0	-1691.2	3392.3	48.49	0.600	0.805	0.602	0.806
(84) $(a_0 + a_1 * (BD^{(1/3)})) + a_2 * (RHIC^{(1/3)})^3$	a0	-1691.3	3392.7	48.56	0.603	0.804	0.606	0.805
(85) $a_0 * BA^{a_1} * RDIC^{a_2} * TH^{a_3} * CL^{a_4} * DBLC^{a_5}$	a0	-1728.7	3473.4	61.66	0.638	0.638	0.644	0.644
(86) $(a_0 * BA^{a_1}) * (RDIC^{(a_2 - 1)}) * (\exp(a_3 * (RHIC^{a_4})))$	a4	-1731.1	3476.2	62.14	0.633	0.633	0.638	0.638
(87) $BA * (a_0 * RHIC + a_1 * (RHIC^2) + a_2) + a_3$	a3	-1732.3	3476.5	58.76	0.620	0.702	0.623	0.705
(88) $BA * (a_0 * RDIC + a_1 * (RDIC^2) + a_2) + a_3$	a3	-1732.3	3476.5	58.76	0.620	0.702	0.623	0.705
(89) ^e $(a_0 * BD^{a_1}) * (RDIC^{(a_2 - 1)}) * (\exp(a_3 * (RDIC^{a_4})))$	a3	-1733.2	3482.4	62.57	0.628	0.628	0.633	0.633
(90) ^f $(a_0 * BD^{a_1}) * (\exp(a_2 * (RDIC^{a_3})))$	a2	-1736.8	3485.7	63.30	0.621	0.621	0.624	0.624
(91) $(a_0 * BA^{a_1}) * (RHIC^{(a_2 - 1)})$	a0	-1738.7	3487.3	63.67	0.618	0.618	0.620	0.620
(92) $BD * (a_0 * RHIC + a_1 * (RHIC^2) + a_2) + a_3$	a3	-1739.3	3490.5	59.27	0.595	0.702	0.598	0.705
(93) $BD * (a_0 * RDIC + a_1 * (RDIC^2) + a_2) + a_3$	a3	-1739.3	3490.5	59.27	0.595	0.702	0.598	0.705
(94) $(a_0 * BD^{a_1}) * (RDIC^{(a_2 - 1)})$	a0	-1739.7	3489.5	63.89	0.615	0.615	0.617	0.617
(95) $(a_0 * BD^{a_1}) * (RDIC^{(a_2 - 1)})$	a0	-1739.7	3489.5	63.89	0.615	0.615	0.617	0.617
(96) $(a_0 * BD^{a_1}) * (\exp(a_2 * (RDIC^{a_3})))$	a2	-1745.3	3504.6	65.04	0.600	0.600	0.604	0.604
(97) $(a_0 * BD^{a_1})$	a0	-1745.5	3500.9	65.07	0.602	0.602	0.603	0.603

Note: "Rand" denotes variables with random effects; LL, log likelihood; AIC, Akaike information criterion; RMSE, root mean square error.

^aEk 1979.

^bMonserud and Marshall 1999.

^cGillespie et al. 1994.

^dMaguire et al. 1998.

^eMaguire and Bennett 1996.

^fXu and Harrington 1998.

Table A3. Models evaluated for goodness of fit to crown foliar mass.

					Adjusted R ²		Generalized R ²		
Crown foliar mass model equation		Rand	LL	AIC	RMSE	Fixed	Fixed + random	Fixed	Fixed + random
(5)	(b1 * DBH) + (b2 * HMC)	b1 + b2	-419.3	850.5	1032.18	0.832	0.934	0.835	0.935
(98) ^a	b0 * (DBH^(b1 + (b2 * CL) + b3 * (TH/DBH)))	b3	-419.4	850.8	1012.20	0.814	0.935	0.825	0.939
(99)	(b0 * (HMC^b1) * DBH^(b2 + b3 * (TH/DBH)))	b3	-419.8	851.7	1047.76	0.818	0.929	0.830	0.933
(100)	(b0 * DBH^(b1 + b2 * (TH/DBH)))	b2	-419.9	849.8	1060.21	0.825	0.928	0.833	0.931
(101)	b0 * (DBH^b1) * (CLR^b2) * (CL^b3)	b3	-420.9	853.8	1100.29	0.821	0.919	0.832	0.924
(102)	(b0 * DBH^(b1 + b2 * (HMC/DBH)))	b2	-420.9	851.8	1105.98	0.825	0.920	0.832	0.923
(103)	b0 * DBH^b1 * HMC^b2	b1	-422.0	854.0	1143.83	0.821	0.913	0.828	0.917
(104)	b0 + (b1 * DBH) + (b2 * MCLR) + (b3 * HMC)	b0	-422.2	856.5	1132.57	0.827	0.916	0.838	0.922
(105)	b0 + (b1 * DBH) + (b2 * HMC)	b0	-422.3	854.5	1132.61	0.830	0.918	0.837	0.922
(106) ^b	(1/HMC) * ((DBH/b1)^b2)	b2	-422.8	853.7	1184.20	0.823	0.908	0.827	0.909
(107)	b0 + (b1 * DBH) + (b2 * MCLR)	b0	-423.7	857.3	1126.76	0.814	0.921	0.822	0.925
(108)	b0 * (DBH^b1) * exp(b2 * (DBH/HMC))	b2	-424.5	858.9	1195.39	0.802	0.905	0.810	0.909
(109)	b0 * DBH^b1 * exp(MCLR * b2)	b1	-425.9	861.8	1159.85	0.779	0.915	0.788	0.919
(110)	b0 * DBH^b1 * MCLR^b2	b1	-426.2	862.3	1132.79	0.772	0.921	0.781	0.924
(111)	b0 * DBH^b1 * exp(HMC * b2)	b1	-426.5	863.0	1186.47	0.790	0.910	0.798	0.914
(112)	b0 * DBH^b1 * CLR^b2	b1	-426.6	863.1	1125.82	0.767	0.922	0.777	0.925
(113)	b0 * DBH^b1 * exp(CLR * b2)	b1	-426.7	863.3	1152.46	0.772	0.917	0.781	0.921
(114)	DBH^b1 * exp(CL * b2)	b1	-426.9	861.7	956.52	0.688	0.949	0.694	0.950
(115) ^c	b0 + b1 * (DBH + MCLR)	b0	-427.0	862.1	1094.91	0.721	0.932	0.727	0.933
(116)	b0 * DBH^b1	b1	-427.9	863.8	1046.51	0.734	0.937	0.740	0.939
(117)	(DBH^b1) * (CLR^b2) * (CL^b3)	b3	-428.4	866.7	1047.17	0.676	0.936	0.690	0.939
(118) ^d	b0 * (DBH^b1) * exp(b2 * (DBH/CL))	b2	-428.5	867.0	1026.00	0.712	0.938	0.724	0.941
(119) ^e	b0 * (CL^b1) * exp(b2 * (DBH/TH))	b2	-428.7	867.5	1335.36	0.772	0.879	0.781	0.884
(120)	DBH^b1 * CL^b2	b1	-429.1	866.1	1043.10	0.679	0.938	0.686	0.940
(121)	DBH^b1	b1	-429.9	865.7	1020.69	0.611	0.943	0.611	0.943
(122)	(DBH^(b1 + b2 * (TH/DBH)))	b2	-430.4	868.7	1032.20	0.626	0.941	0.634	0.943

Note: "Rand" denotes variables with random effects; LL, log likelihood; AIC, Akaike information criterion; RMSE, root mean square error.

^aWeiskittel et al. 2009.

^bJerez et al. 2005.

^cKenefic and Seymour 1999.

^dNelson et al. 2014.

^eMaguire and Bennett 1996.

Table A4. Models evaluated for goodness of fit to Weibull kurtosis parameter estimates.

Weibull kurtosis model equation					Adjusted R ²		Generalized R ²		
					Fixed	Fixed + random	Fixed	Fixed + random	
(123)	exp(g0 + (CL * g1) + (DBLC * g2) + (CFM * g3))	g1	72.6	-133.3	0.0549	0.546	0.546	0.574	0.574
(6)	g0 + g1 * CL + g2 * DBLC	NA	70.5	-133.0	0.0592	0.516	NA	0.536	NA
(124)	g0 + (CL * g1) + (DBLC * g2)	g1	70.5	-131.0	0.0574	0.516	0.516	0.536	0.536
(125)	g0 * (CL^(g1 + (DBLC * g2) + (CFM * g3)))	g1	70.0	-128.1	0.0579	0.495	0.495	0.527	0.527
(126)	g0 * (CL^(g1 + (g2 * DBLC)))	g1	68.8	-127.6	0.0594	0.480	0.480	0.502	0.502
(127)	g0 * (CL^g1) * (DBLC^g2)	g1	68.1	-126.2	0.0592	0.464	0.502	0.486	0.523
(128)	exp(g0 + (CL * g1) + (DBH * g2) + (CFM * g3))	g1	64.2	-116.4	0.0653	0.359	0.359	0.399	0.399
(129)	g0 + g1 * CL + g2 * DBH + g3 * CFM	NA	64.1	-118.3	0.0682	0.357	NA	0.397	NA
(130)	g0 + g1 * CL + g2 * DBH	NA	64.0	-119.9	0.0677	0.367	NA	0.394	NA
(131)	g0 * (CL^g1) * exp(DBH * g2)	g1	63.5	-117.1	0.0662	0.356	0.356	0.383	0.383
(132)	g0 * (CL^(g1 + (DBH * g2)))	g1	63.5	-117.0	0.0662	0.354	0.354	0.381	0.381
(133)	g0 * (CL^g1) * (DBH^g2)	g1	63.4	-116.8	0.0663	0.353	0.353	0.379	0.379
(134)	g0 * (CL^g1) * (CFM^g2)	g1	63.3	-116.7	0.0664	0.351	0.351	0.378	0.378
(135)	g0 * (CL^(g1 + (g2 * CL)))	g1	62.6	-115.1	0.0675	0.330	0.330	0.358	0.358
(136)	g0 * (CL^g1) * (TH^g2)	g1	62.5	-115.0	0.0675	0.329	0.329	0.357	0.357
(137)	g0 * (CL^g1) * (HMC^g2)	g1	62.5	-115.0	0.0676	0.328	0.328	0.356	0.356
(138)	g0 + (CL * g1)	g1	62.4	-116.9	0.0677	0.340	0.340	0.354	0.354
(139)	g0 + CL * g1	NA	62.4	-118.9	0.0691	0.340	NA	0.354	NA
(140)	g0 * (CL^g1)	NA	62.3	-118.6	0.0693	0.337	NA	0.350	NA
(141)	g0 * (CL^g1)	g1	62.3	-116.6	0.0679	0.337	0.337	0.350	0.350
(142)	CL^(g1 + (DBH * g2) + (CFM * g3))	g1	58.4	-106.9	0.0716	0.204	0.281	0.237	0.311
(143)	g0 * (TH^g1)	g1	57.3	-106.5	0.0726	0.179	0.288	0.196	0.303
(144)	g0 + TH * g1	NA	57.2	-108.4	0.0769	0.183	NA	0.200	NA
(145)	CL^(g1 + (g2 * DBLC))	g1	57.1	-106.2	0.0704	0.156	0.364	0.174	0.378
(146)	CL^(g1 + (g2 * DBLC))	NA	56.7	-107.4	0.0777	0.166	NA	0.184	NA
(147)	g0 * (DBH^g1)	g1	56.6	-105.2	0.0734	0.157	0.276	0.175	0.291
(148)	g0 * (HMC^g1)	g1	55.9	-103.7	0.0727	0.117	0.315	0.135	0.330

Note: "Rand" denotes variables with random effects; LL, log likelihood; AIC, Akaike information criterion; RMSE, root mean square error; NA, model did not include random effects.

Table A5. Models evaluated for goodness of fit to Weibull skew parameter estimates.

Weibull skew model equation		Rand	LL	AIC	RMSE	Adjusted R ²		Generalized R ²	
						Fixed	Fixed + random	Fixed	Fixed + random
(149)	$g3 * (CLR^{g4} + (CFM * g5))$	$g4 + g5$	-36.7	87.3	0.387	0.411	0.852	0.436	0.858
(150)	$g3 * (CLR^{g4} + (CFM * g5) + (HBLC * g6))$	$g4 + g6$	-37.9	91.7	0.459	0.236	0.763	0.284	0.778
(7)	$g3 * (CLR^{g4} + (CFM * g5))$	$g4$	-41.2	92.4	0.497	0.522	0.731	0.542	0.742
(151)	$g3 + (CL * g4) + (g5 * HBLC) + (g6 * CL * HBLC)$	$g5 + g6$	-41.8	99.6	0.483	0.384	0.749	0.423	0.765
(152)	$g3 * (CLR^{g4}) * (CFM^{g5})$	$g4 + g5$	-41.8	97.7	0.505	0.487	0.718	0.508	0.730
(153)	$g3 * (CLR^{g4}) * (CFM^{g5})$	$g4$	-42.5	95.1	0.515	0.496	0.710	0.517	0.722
(154)	$g3 * (CLR^{g4} + (CFM * g5)) * (CFM^{g6})$	$g5$	-42.8	97.6	0.535	0.501	0.669	0.532	0.690
(155)	$g3 * (CLR^{g4} + (CFM * g5))$	$g5$	-43.0	95.9	0.539	0.514	0.669	0.534	0.683
(156)	$g3 * (CLR^{g4})$	$g4$	-45.2	98.4	0.541	0.418	0.687	0.430	0.694
(157)	$g3 + (CLR * g4) + (g5 * HBLC)$	$g5$	-48.4	106.9	0.616	0.447	0.559	0.470	0.578
(158)	$g3 * (CFM^{g4}) * (HBLC^{g5})$	$g5$	-48.8	107.6	0.641	0.456	0.500	0.478	0.521
(159) ^a	$g3 * (CL^{g4} + (CFM * g5) + (TH * g6))$	$g4 + g6$	-49.6	115.3	0.602	0.373	0.592	0.412	0.617
(160)	$g3 + (CFM * g4) + (g5 * HBLC)$	$g3$	-49.9	109.8	0.660	0.431	0.464	0.455	0.486
(161)	$g3 * HBLC^{g4} + (g5 * CL)$	$g4$	-50.1	110.3	0.619	0.403	0.577	0.428	0.594
(162)	$g3 + CLR * g4$	NA	-50.8	107.6	0.697	0.423	NA	0.435	NA
(163)	$g3 * (CL^{g4}) * (HBLC^{g5})$	$g4$	-51.6	113.3	0.673	0.387	0.459	0.412	0.481
(164)	$g3 + (CL * g4) + (g5 * HBLC)$	$g3$	-52.9	115.8	0.667	0.337	0.497	0.364	0.518
(165)	$g3 * (HBLC^{g4})$	$g4$	-56.9	121.7	0.719	0.232	0.431	0.248	0.443

Note: “Rand” denotes variables with random effects; LL, log likelihood; AIC, Akaike information criterion; RMSE, root mean square error; NA, model did not include random effects.

^aWeiskittel et al. 2009.

Appendix references

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